Adil Karjauv

Contact Information

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Research Background

- **Deep Learning**: Efficient video generative AI, efficient video processing, adversarial machine learning and its applications in multimedia.

Education

KAIST, Daejeon, Republic of Korea — 2019–2021

- M.S. in Electrical Engineering (Advisor: In So Kweon)

KAIST, Daejeon, Republic of Korea — 2014–2019

- B.S. in Computer Science

Research Experience

- **Qualcomm AI Research** — Amsterdam, Netherlands
  Machine Learning Researcher (Senior Engineer) — Dec. 2025 - Present
  Machine Learning Researcher (Engineer) — Oct. 2022 - Dec. 2025
  - **MobileWan:** Led an architecture optimization effort for the largest video generation diffusion transformer running on-device (based on Wan2.2-5B), developed a novel structured pruning method, improving efficiency without sacrificing quality. Paper under review.
  - **PyramidalWan:** Co-developed a step distillation method for a pyramidally-distilled video generation model, achieving a 10x inference speedup. Published at CVPR 2026; patent filed.
  - **Neodragon:** Contributed to step distillation techniques (2.5-10x speedup) for the first diffusion transformer based mobile video generation model. Presented as a NeurIPS 2025 demo and published at ICLR 2026; patent to be filed.
  - **MoViE:** Developed and optimized a state-of-the-art diffusion model for video editing, achieving an 87x on-device speedup – the fastest diffusion-based video editing demo shown on-device. Presented at NeurIPS 2024 and CVPR 2025 as a demo.
  - **Object-Centric Diffusion (OCD):** Contributed to training-free techniques that reallocate compute toward salient regions in video editing, cutting latency up to 10x with comparable quality. Published at ECCV 2024.
  - Pioneered the first on-device deep-learning-based video denoising solution at high resolution (QHD/4K at 30FPS); patent filed.
- **Robotics and Computer Vision Lab** — Daejeon, Republic of Korea
  Graduate Student Researcher — Sept. 2019 - Jan. 2022
  - Conducted research on machine learning and computer vision with a focus on adversarial machine learning and its applications in multimedia. Published extensively in top machine learning venues (conferences and workshops).
- **Crazing Lab** — Seoul, Republic of Korea
  Research Intern — Apr. 2019 - July 2019
  - **Robotic Vision System:** Developed a robotic vision system based on several CSI cameras and NVIDIA embedded computer (Jetson Xavier). Utilized a low-level camera API and GStreamer framework to configure cameras. Designed and implemented algorithms for a real-time cylindrical panorama projection for the system.
    Technologies: Python, C++, NumPy, OpenCV, ROS, Libargus Camera API, GStreamer.
- **Motion Computing Lab** — Daejeon, Republic of Korea
  Research Intern — June 2018 - Aug. 2018
  - **Data-Driven Cloth Simulation:** Implemented scripts that automated data generation for model training by performing motion retargeting using computer graphics software API.
    Technologies: Python, Autodesk Maya, Autodesk MotionBuilder.
- **Computational Media Lab** — Daejeon, Republic of Korea
  Research Intern — June 2017 - Sep. 2018
  - **MotionSnap:** Conducted research on visual media retrieval using motion and other sensors data embedded in mobile and wearable devices. Designed a new approach for automatic photo capture and video editing and implemented algorithms based on the approach for many use cases such as jumping pictures retrieval, automatic camera selection, etc. Developed several Android applications based on these algorithms (see demo video for more details).
    Technologies: Java, Python, NumPy, Android Studio, OpenCV.
    Demo video: http://bit.ly/motionsnap_demo.

International Conferences

1. Mohsen Ghafoorian*, Denis Korzhenkov*, **Adil Karjauv***, ..., (*Equal contribution), "MobileWan: Closing the Quality Gap for Mobile Video Diffusion", **arXiv preprint**.
2. Denis Korzhenkov*, **Adil Karjauv***, Animesh Karnewar, Mohsen Ghafoorian, Amirhossein Habibian (*Equal contribution), "PyramidalWan: On Making Pretrained Video Model Pyramidal for Efficient Inference", in **CVPR 2026**.
3. Animesh Karnewar, Denis Korzhenkov, Ioannis Lelekas, Noor Fathima, **Adil Karjauv**, ..., "Neodragon: Mobile Video Generation using Diffusion Transformer", in **ICLR 2026**.
4. **Adil Karjauv***, Noor Fathima*, Ioannis Lelekas, Fatih Porikli, Amir Ghodrati, Amirhossein Habibian (*Equal contribution), "MoViE: Mobile Diffusion for Video Editing", **arXiv preprint**.
5. Kumara Kahatapitiya, **Adil Karjauv**, Davide Abati, Fatih Porikli, Yuki M. Asano, Amirhossein Habibian, "Object-Centric Diffusion for Efficient Video Editing", in **ECCV 2024**.
6. Chaoning Zhang, Philipp Benz, **Adil Karjauv**, In So Kweon, Choong Seon Hong, "Simple Techniques are Sufficient for Boosting Adversarial Transferability", in **ACM MM 2023**.
7. Chaoning Zhang, Philipp Benz, **Adil Karjauv**, Jae Won Cho, Kang Zhang, In So Kweon, "Investigating Top-k White-Box and Transferable Black-box Attack", in **CVPR 2022**.
8. Philipp Benz*, Soomin Ham*, Chaoning Zhang*, **Adil Karjauv**, In So Kweon (*Equal contribution), "Adversarial Robustness Comparison of Vision Transformer and MLP-Mixer to CNNs", in **BMVC 2021**.
9. Chaoning Zhang*, **Adil Karjauv***, Philipp Benz*, In So Kweon (*Equal contribution), "Towards Robust Deep Hiding Under Non-Differentiable Distortions for Practical Blind Watermarking", in **ACM MM 2021**.
10. Chaoning Zhang*, Philipp Benz*, **Adil Karjauv***, In So Kweon (*Equal contribution), "Data-free Universal Adversarial Perturbation and Black-box Attack", in **ICCV 2021**.
11. Chaoning Zhang*, Philipp Benz*, Chenguo Lin*, **Adil Karjauv**, Jing Wu, In So Kweon (*Equal contribution), "A Survey on Universal Adversarial Attack", in **IJCAI 2021**.
12. Chaoning Zhang*, Philipp Benz*, **Adil Karjauv**, In So Kweon (*Equal contribution), "Universal Adversarial Training With Class-Wise Perturbations", in **ICME 2021**.
13. **Adil Karjauv***, Sanzhar Bakhtiyarov*, Chaoning Zhang, Jean-Charles Bazin, In So Kweon (*Equal contribution), "MotionSnap: A Motion Sensor-Based Approach for Automatic Capture and Editing of Photos and Videos on Smartphones", in **ICME 2021**.
14. Chaoning Zhang*, Philipp Benz*, **Adil Karjauv**, In So Kweon (*Equal contribution), "Universal Adversarial Perturbations Through the Lens of Deep Steganography: Towards a Fourier Perspective", in **AAAI 2021**.
15. Chaoning Zhang*, Philipp Benz*, **Adil Karjauv**, In So Kweon (*Equal contribution), "Revisiting Batch Normalization for Improving Corruption Robustness", in **WACV 2021**.
16. Chaoning Zhang*, Philipp Benz*, **Adil Karjauv***, Geng Sun, In So Kweon (*Equal contribution), "UDH: Universal Deep Hiding for Steganography, Watermarking, and Light Field Messaging", in **NeurIPS 2020**.

Selected Workshops

1. Elia Peruzzo, **Adil Karjauv**, Nicu Sebe, Amir Ghodrati, Amirhossein Habibian, "AdapToR: Adaptive Token Reduction for Video Diffusion Transformers", in **CVPR 2025 Workshop**.
2. Chaoning Zhang*, **Adil Karjauv***, Philipp Benz*, Soomin Ham, Gyusang Cho, Chan-Hyun Youn, In So Kweon (*Equal contribution), "Is FGSM Optimal or Necessary for L_∞ Adversarial Attack?", in **CVPR 2021 AML-CV Workshop**.
3. Chaoning Zhang*, Philipp Benz*, **Adil Karjauv***, In So Kweon (*Equal contribution), "On Strength and Transferability of Adversarial Examples: Stronger Attack Transfers Better", in **ICLR 2021 RobustML Workshop**.
4. Chaoning Zhang*, Philipp Benz*, **Adil Karjauv***, In So Kweon (*Equal contribution), "Stochastic Depth Boosts Transferability of Non-targeted and Targeted Adversarial Attacks", in **ICLR 2021 RobustML Workshop**.

5. Chaoning Zhang*, Philipp Benz*, **Adil Karjauv***, Jae Won Cho, In So Kweon (*Equal contribution), "Towards Data-free Universal Adversarial Perturbations with Artificial Jigsaw Images", in **ICLR 2021 RobustML Workshop**.

Patents

- Adversarial Distillation of Pyramidal Flow Matching Model (pending, 63/884,922)
- Pyramidal Distillation of Video Diffusion Transformers (pending, 19/673,610)
- AdapToR: Adaptive Token Reduction for Video Diffusion Transformers (pending, 19/257,306)
- Guidance Oscillation for Multi-Modal Diffusion Models (pending, 18/974,441)
- Multimodal Guidance Distillation for Efficient Diffusion Models (pending, 18/770,606)
- Noise Injection for Generalized Blind Denoising (pending, 18/589,018)

Honors and Awards

- Outstanding Paper Award in CVPR 2021 AML-CV Workshop — 2021
- 9th Place Award (out of 1681 teams) CVPR 2021 Security AI Challenger Phase VI — 2021
- URP (Undergraduate Research Program) Best Poster Award — 2018
- Excellence Prize in the E*5 Competition by Startup KAIST (7,000$ Cash Award) — 2018
- KAIST Undergraduate Scholarship: Full Tuition Fee and Monthly Stipend — 2014 - 2019

Technical Skills

- **Programming languages:** Python, Java, C++, C, MATLAB.
- **Technologies:** PyTorch, DeepSpeed, Diffusers, TensorFlow, TensorBoard, Keras, NumPy, OpenCV, Git, SQL, ROS.
- **Skills:** Deep Learning, Generative Modeling, Diffusion Models, Efficient ML (Distillation/Pruning), Distributed Training (DDP/FSDP/DeepSpeed), On-Device Inference, Multimodal & Foundation Models, Adversarial ML.